

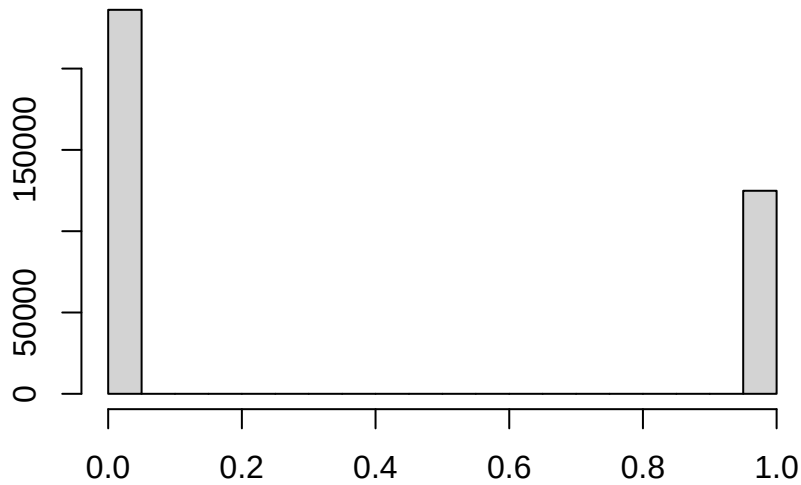
Competition – Contractor Experience

2024-06-03

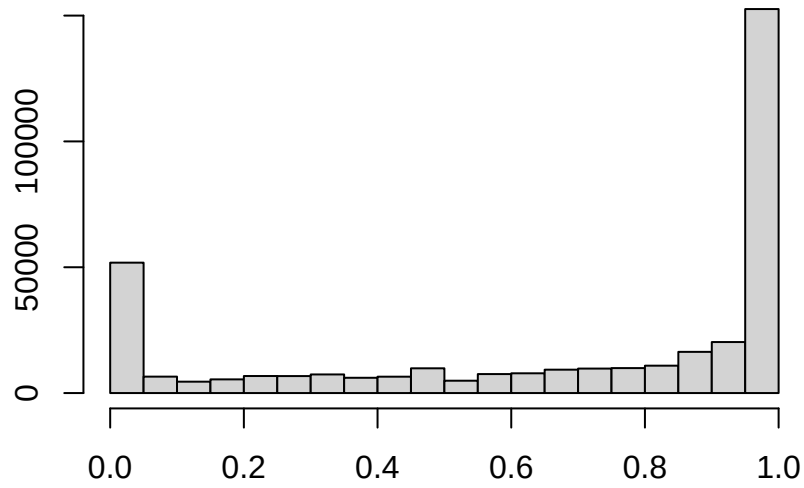
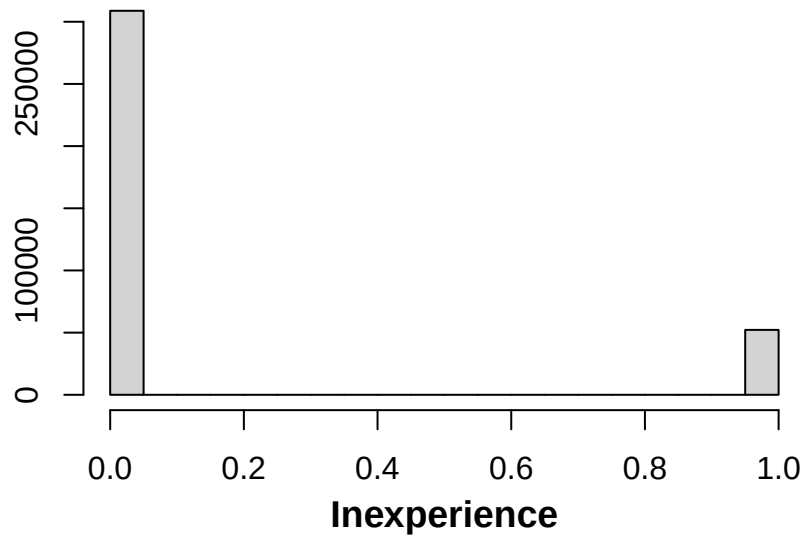
1 Time Invariant Status

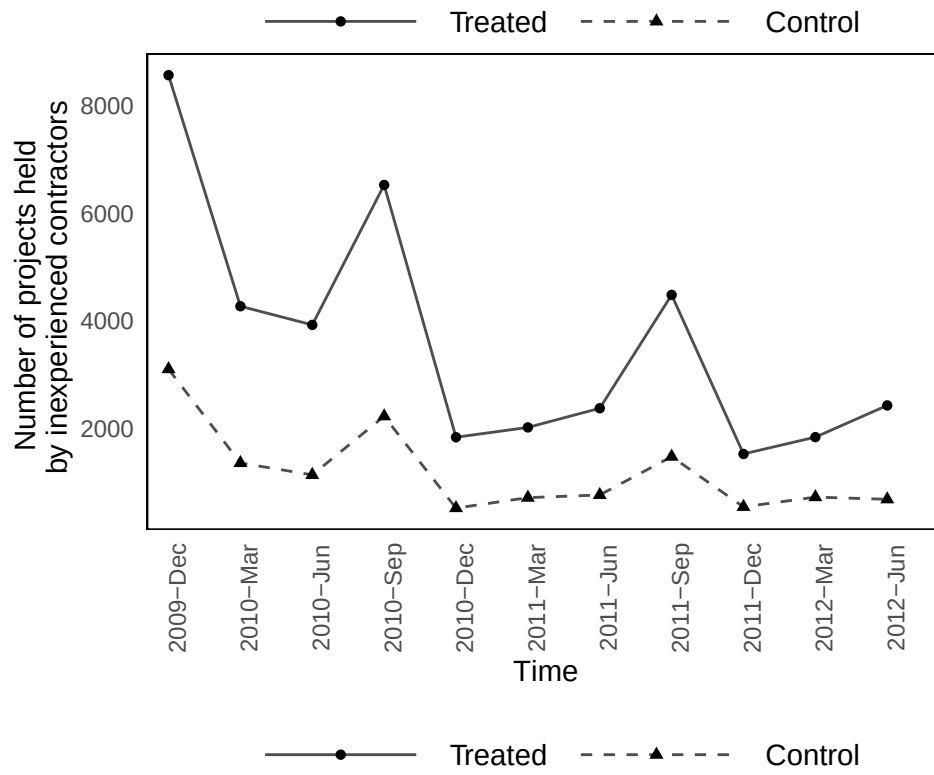
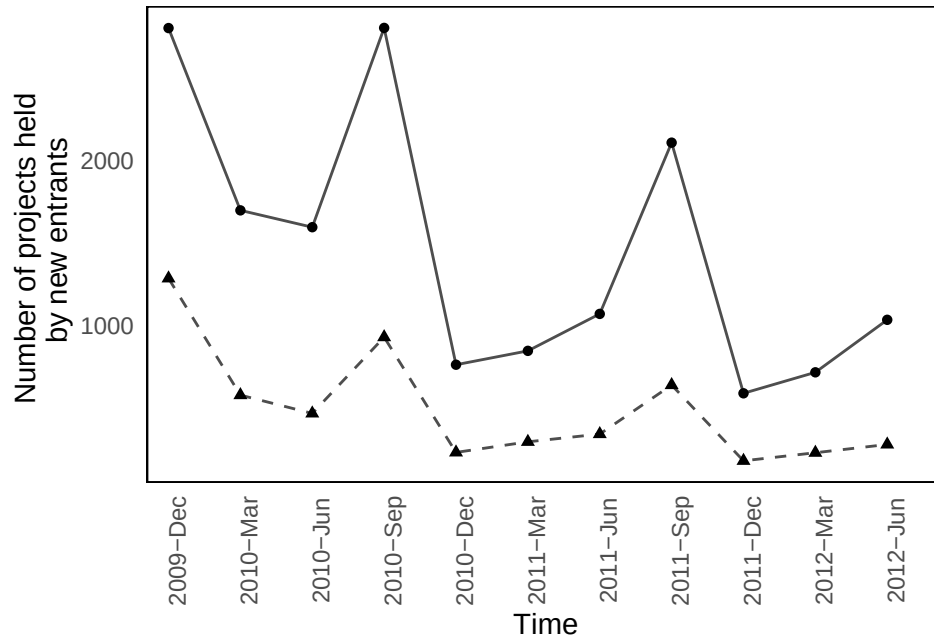
- Let n_i^{task} = Number of projects of the same task held by the contractor in the three quarters before starting project i
- Let n_i = Number of projects held by the contractor in the three quarters before starting project i
- Define:
 - Inexperienced contractor $_i$ = 1 if $n_i^{task} = 0$
 - New entrant $_i$ = 1 if $n_i = 0$
 - Level of inexperience $_i$ = $1 - \frac{n_i^{task}}{n_i}$ if $n_i > 0$
 - Level of inexperience $_i$ = 1 if $n_{i,k,t} = 0$.

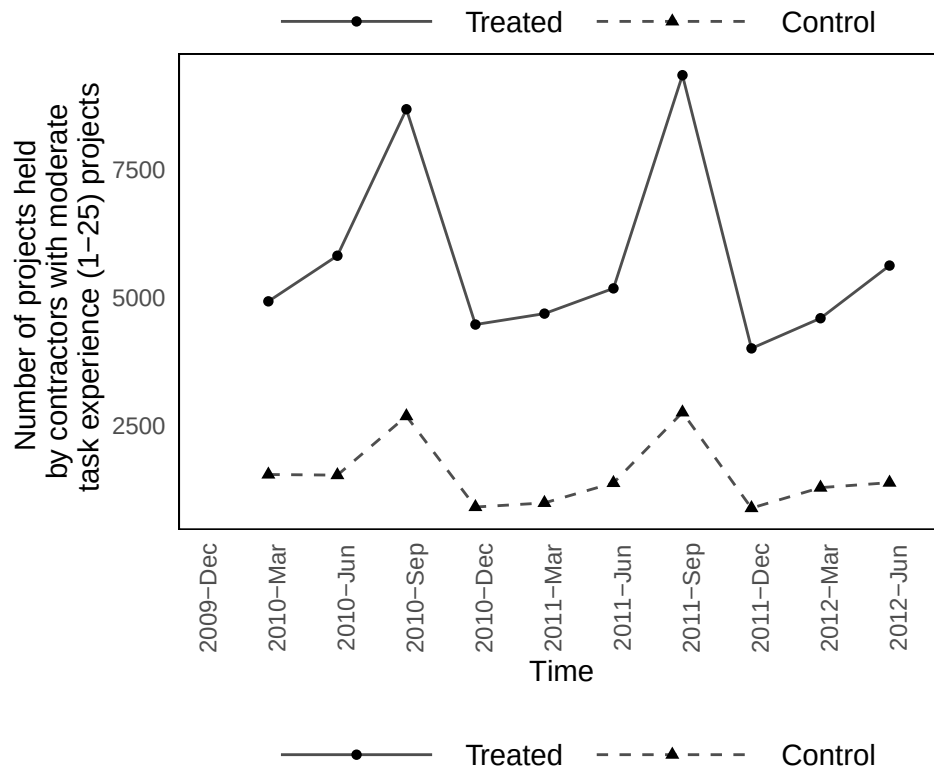
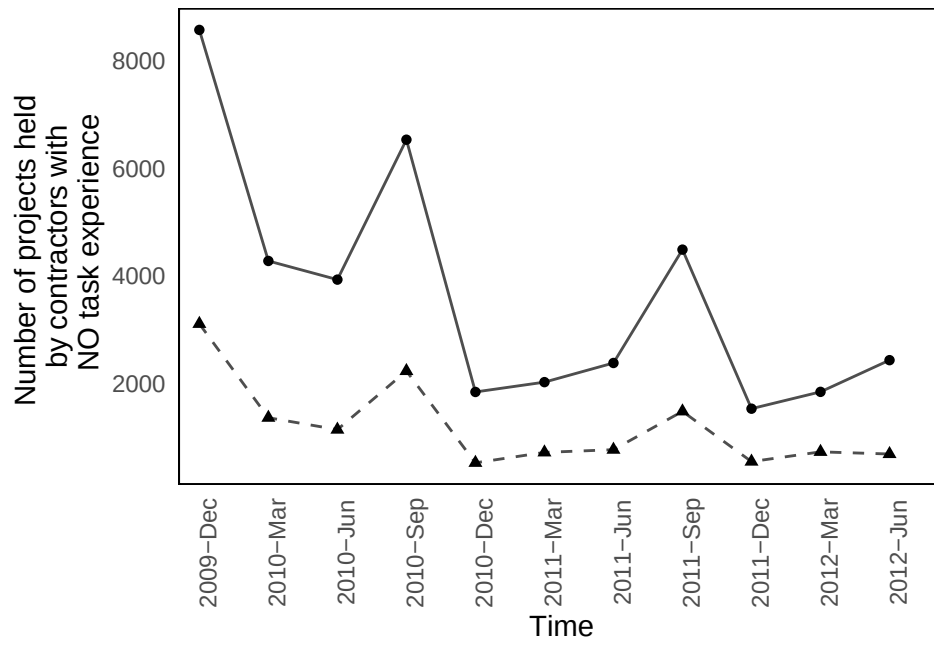
Task inexperience

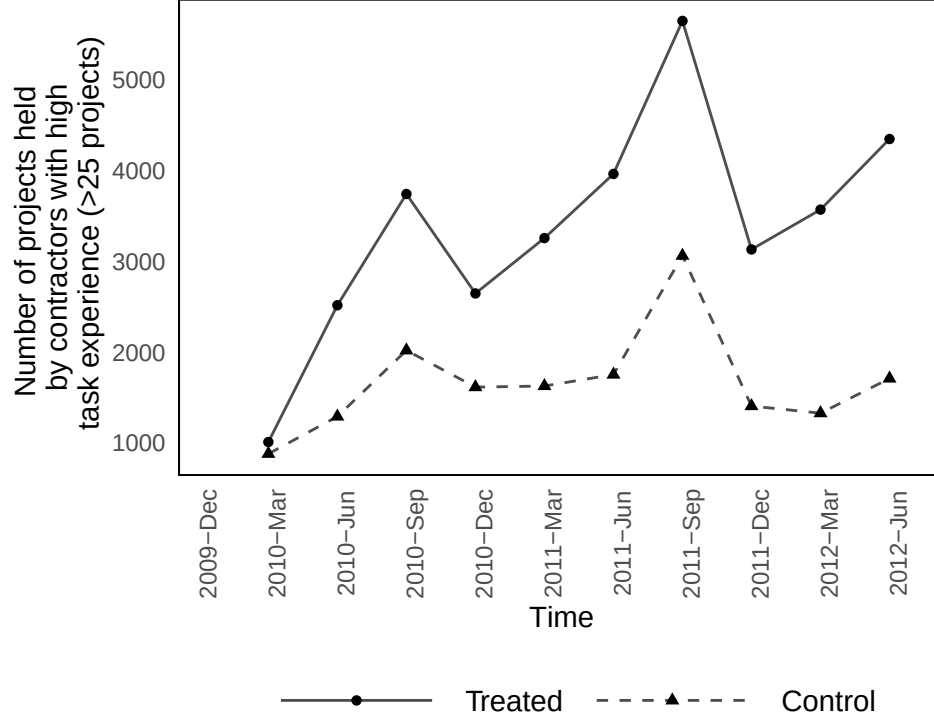


New Entrant









2 Regression results

Table 1: Logit model

	Inexperienced contractor			
	I	II	III	IV
Constant	0.181*** (0.023)			
Treat	0.066*** (0.024)	0.088*** (0.025)	0.262*** (0.034)	0.225*** (0.035)
Ln(Stage)	0.140*** (0.005)	0.180*** (0.005)	0.241*** (0.006)	0.243*** (0.006)
Post	-0.771*** (0.022)			
Treat × Post	-0.064** (0.026)	-0.081*** (0.026)	0.048 (0.031)	0.069** (0.031)
Time FE	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes
Industry FE	No	No	No	Yes
Observations	184,976	178,669	175,095	175,078
Squared Correlation	0.04	0.04	0.25	0.27
Pseudo R ²	0.03	0.03	0.21	0.22
BIC	240,067.46	228,585.31	187,362.22	185,863.22

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Percentage delay rate					
	I	II	III	IV	V	VI
Constant	5.381*** (0.151)	9.934*** (0.161)				
Treat	-2.638*** (0.170)	-2.423*** (0.164)	-2.441*** (0.164)	-0.367** (0.163)	-0.419*** (0.162)	-0.497*** (0.151)
Inexperienced contractor	0.203 (0.221)	-0.204 (0.210)	-0.121 (0.211)	-0.411** (0.200)	-0.418** (0.200)	1.278*** (0.183)
Post	0.021 (0.179)	-0.955*** (0.173)				
Treat × Inexperienced contractor	0.748*** (0.251)	0.958*** (0.239)	0.953*** (0.239)	-0.264 (0.224)	-0.197 (0.224)	-0.552*** (0.206)
Inexperienced contractor × Post	0.844*** (0.281)	0.725*** (0.269)	0.655** (0.270)	0.045 (0.262)	0.003 (0.261)	0.094 (0.255)
Treat × Post	0.820*** (0.203)	0.967*** (0.196)	0.992*** (0.196)	0.636*** (0.185)	0.636*** (0.185)	0.798*** (0.177)
Treat × Inexperienced contractor × Post	-0.020 (0.324)	-0.475 (0.312)	-0.479 (0.312)	-0.242 (0.301)	-0.206 (0.301)	0.377 (0.301)
Ln(Stage)		2.829*** (0.030)	2.870*** (0.031)	2.770*** (0.030)	2.748*** (0.030)	2.666*** (0.030)
Num offers						-0.006 (0.008)
Ln(Initial Budget)						0.429*** (0.021)
Ln(Initial Duration)						-7.400*** (0.092)
Time FE	No	No	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes
Observations	184,908	184,882	184,882	184,882	184,882	165,473
R ²	0.01	0.06	0.06	0.12	0.12	0.22
Within R ²			0.06	0.05	0.05	0.15

Clustered (Project ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Logit model

	New Entrant				
	I	II	III	IV	V
Constant	-1.503*** (0.021)	-1.164*** (0.030)			
Treat	-0.076*** (0.024)	-0.053 (0.032)	-0.048 (0.033)	0.297*** (0.043)	0.304*** (0.044)
Post	-0.512*** (0.024)	-0.520*** (0.029)			
Treat $\times$ Post	0.104*** (0.028)	0.120*** (0.034)	0.118*** (0.034)	0.158*** (0.038)	0.174*** (0.038)
Ln(Stage)		0.114*** (0.007)	0.133*** (0.007)	0.149*** (0.008)	0.149*** (0.008)
Time FE	No	No	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes
Industry FE	No	No	No	No	Yes
Observations	361,013	184,976	184,976	182,001	181,977
Squared Correlation	0.01	0.01	0.02	0.18	0.19
Pseudo R ²	0.01	0.01	0.02	0.18	0.20
BIC	296,200.67	165,117.57	163,976.91	141,579.17	140,278.91

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

	Percentage delay rate					
	I	II	III	IV	V	VI
Constant	5.472*** (0.123)	9.929*** (0.136)				
Treat	-2.474*** (0.139)	-2.244*** (0.132)	-2.257*** (0.132)	-0.558*** (0.136)	-0.575*** (0.136)	-0.624*** (0.126)
New Entrant	0.044 (0.280)	-0.369 (0.265)	-0.336 (0.265)	-0.042 (0.256)	-0.036 (0.257)	1.222*** (0.228)
Post	0.084 (0.149)	-0.880*** (0.144)				
Treat $\times$ New Entrant	1.090*** (0.324)	1.509*** (0.307)	1.483*** (0.306)	0.102 (0.294)	0.133 (0.295)	-0.710*** (0.271)
New Entrant $\times$ Post	1.254*** (0.371)	1.166*** (0.355)	1.136*** (0.355)	0.269 (0.346)	0.211 (0.346)	0.185 (0.337)
Treat $\times$ Post	0.799*** (0.170)	0.886*** (0.164)	0.904*** (0.164)	0.699*** (0.157)	0.690*** (0.157)	0.896*** (0.152)
Treat $\times$ New Entrant $\times$ Post	-0.570 (0.431)	-1.172*** (0.415)	-1.151*** (0.414)	-0.601 (0.404)	-0.550 (0.404)	0.434 (0.409)
Ln(Stage)		2.840*** (0.030)	2.885*** (0.031)	2.746*** (0.030)	2.725*** (0.030)	2.694*** (0.030)
Num offers						-0.006 (0.008)
Ln(Initial Budget)						0.435*** (0.021)
Ln(Initial Duration)						-7.366*** (0.092)
Time FE	No	No	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes
Observations	184,908	184,882	184,882	184,882	184,882	165,473
R ²	0.01	0.06	0.06	0.12	0.12	0.22
Within R ²			0.06	0.05	0.05	0.15

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

3 Regressions – Continuous variable

	Level of inexperience			
	I	II	III	IV
Constant	0.797*** (0.004)			
Treat	0.029*** (0.004)	0.033*** (0.004)	-0.004 (0.004)	-0.007* (0.004)
Ln(Stage)	0.018*** (0.001)	0.021*** (0.001)	0.025*** (0.001)	0.025*** (0.001)
Post	-0.147*** (0.004)			
Treat × Post	0.015*** (0.004)	0.012*** (0.004)	0.031*** (0.004)	0.032*** (0.004)
Time FE	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes
Industry FE	No	No	No	Yes
Observations	184,976	184,976	184,976	184,976
R ²	0.04	0.06	0.27	0.27
Within R ²		0.01	0.01	0.01

Clustered (Project ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Percentage delay rate					
	I	II	III	IV	V	VI
Constant	6.369*** (0.286)	11.066*** (0.282)				
Treat	-2.594*** (0.333)	-2.327*** (0.317)	-2.373*** (0.317)	0.222 (0.304)	0.121 (0.303)	-0.310 (0.283)
Level of inexperience	-1.156*** (0.336)	-1.525*** (0.320)	-1.564*** (0.321)	-0.765** (0.307)	-0.744** (0.306)	1.071*** (0.282)
Post	-0.850*** (0.327)	-1.800*** (0.313)				
Treat × Level of inexperience	0.461 (0.389)	0.540 (0.370)	0.582 (0.370)	-0.938*** (0.349)	-0.833** (0.347)	-0.507 (0.324)
Level of inexperience × Post	1.505*** (0.398)	1.409*** (0.380)	1.460*** (0.381)	0.854** (0.364)	0.773** (0.365)	0.593* (0.354)
Treat × Post	1.356*** (0.383)	1.582*** (0.367)	1.633*** (0.368)	0.884** (0.346)	0.896*** (0.345)	1.055*** (0.334)
Treat × Level of inexperience × Post	-1.010** (0.465)	-1.361*** (0.445)	-1.404*** (0.445)	-0.629 (0.423)	-0.623 (0.422)	-0.258 (0.414)
Ln(Stage)		2.868*** (0.031)	2.917*** (0.031)	2.771*** (0.030)	2.749*** (0.030)	2.687*** (0.030)
Num offers						-0.008 (0.008)
Ln(Initial Budget)						0.435*** (0.021)
Ln(Initial Duration)						-7.381*** (0.092)
Time FE	No	No	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes
Observations	184,908	184,882	184,882	184,882	184,882	165,473
R ²	0.00	0.06	0.06	0.12	0.12	0.22
Within R ²			0.06	0.05	0.05	0.15

Clustered (Project ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*